

OUR WATER SUPPLY

Objective: To understand the water cycle and the possible contaminants and their sources in our water supply.

To demonstrate your learning, you will construct a diagram of a town (you can make one up), that illustrates the water cycle and contaminants entering the cycle from sources in the town.

Complete the following steps, take your own notes as you will be responsible for this information on tests. You may use valid internet sources (some are provided) and your textbook for information. (p 291-294 and 337-340)

HINT: some of the pollutant information could be presented in a table, be sure to identify the question answered by each column though!

Step 1: Understanding the water cycle

a. You must include the following terms. Make sure you understand them (do NOT include definitions in assignment)!

ground aquifer

ground water

surface water

condensation, precipitation and evaporation

water vapour

b. Draw a rough sketch of the water cycle (showing above terms correctly) for your notes.

Website: [MetLink - Royal Meteorological Society The Changing Water Cycle -](#)

Step 2: Understanding the types of contaminants in our water supply (p291-294)

- In one line each:** Define and give two examples for each of- physical, biological and chemical contaminants
- For chemical contaminants: **list** two that biodegrade quickly and two that do not (that stay in our environment for years). **HINT:** check below for other questions to answer about these contaminants while researching them.
- In one to two lines:** Can all contaminants be easily removed? If not, identify a physical or chemical property that inhibits removal?
- List** the Maximum Acceptable Concentrations (MAC) of at least 3 different contaminants
- In one line,** explain what is meant by a “part per million”.

Useful websites:

- www.ec.gc.ca – Choose the Water link from the left-hand list of topics
- <http://water.epa.gov/drink/contaminants/index.cfm> - this is the US site, lots of great

information but be sure to focus on Canadian information.

Step 3: Understanding the sources of contamination

- a. **In two to three lines** (or in table), identify 2 or 3 sources for each of the main types of contamination (physical, biological and chemical)
- b. **In two to three lines**, identify sources for the four specific chemical contaminants identified in Step 2 (c).

Step 4: Understanding the health effects of contamination

- a. **In two to three lines**, describe several possible health effects of biological contaminants.
- b. **In two lines**, describe what happened in Walkerton, On?
- c. For the four chemical contaminants you listed in Step 2 (c), **identify** the health effects they can have (in a table column or in 2-3 lines)
- d. If not answered in above question, **in one to two lines**, what are the health effects of heavy metals such as mercury and cadmium?

Useful website:

<http://www.hc-sc.gc.ca/ewh-semt/water-eau/drink-potab/guide/index-eng.php>

Step 5: Purification of water

Usually, clear water is needed for disinfection, therefore usually one method is used to clear the water and another is used to remove biological and chemical contaminants.

- a. In **one to three words**, identify the type of dispersion that is being treated during primary wastewater treatment.
- b. **In one to two lines**, identify 2 specific and different physical processes that are likely to be used during primary treatment.
- c. In **one to three words**, identify the type of dispersion that is being treated during tertiary wastewater treatment.
- d. In **three to four lines**: Describe at least **3 methods** of removing **biological and/or chemical** contaminants (name of method, what it removes and simple explanation of how it works). Check your textbook for a good start on this!
- e. Despite the fact that wastewater treatments remove 90-95% of pollutants, 2 serious water pollution problems still exist. **In one to three lines**, identify these 2 problems.

Step 6: YOUR TOWN

Link your learning together!

- Draw a hypothetical town **near a body of water (show this!)**
- Show the water cycle;
- Include the names of at least four pollutants you have identified next to the places they come from as they **enter the water cycle** (for example, a town might have a pulp mill or dump that leaks particular chemicals, draw and label the mill and write the contaminant next to it);
- the potential human and/or environmental impacts of these contaminants (list

these impacts next to the names of the pollutants)

You will be sharing your hypothetical town with your classmates 😊